

USING EXCEL FOR DATA ANALYSIS

- Basic Excel (II) -

Formulas and Functions

- Learn to use functions and formulas.
 1. Understanding the anatomy of [Excel Formulas](#), and [what their components mean](#).
 2. Learn basic math functions
 3. Learn basic statistical functions
 4. How to [install an Add-in](#)
 5. How to use the Statistics Add-in
 6. Learn to use [SUMIF](#) / [COUNTIF](#) and [SUMIFS](#) / [COUNTIFS](#) to add cells only when certain conditions are met.
 7. Learn to use [MAXIFS](#), [MINIFS](#), [AVERAGEIF](#)/[AVERAGEIFS](#)
 8. [Nesting](#) Excel functions
 9. Learn to build standalone logical [IF](#) functions and make them more complex by [nesting AND and OR](#) within them.
 10. Learn to use the [SUBTOTAL](#) function
 11. How to use [STOCKHISTORY](#) function to obtain stock data
 12. Text functions - learn to break apart text with the [LEFT](#), [RIGHT](#), [MID](#), [FIND](#) and [LEN](#) functions.
 13. New text functions [TEXTBEFORE](#) and [TEXTAFTER](#)
 14. Learn to use [VLOOKUP](#)/[HLOOKUP](#)/[XLOOKUP](#) to find an approximate/exact match and return it or the corresponding value from another column.
 15. Time/Date related functions
 16. Unique Registers
 17. Learn to work with formula auditing tools including [tracing precedent and dependent cells](#).



Basic Excel (II) - 1

- Learn to use functions and formulas.
 - Understanding the anatomy of [Excel Formulas](#), and [what their components mean](#).

<https://www.youtube.com/watch?v=Jl0Qk63z2ZY>





Basic Excel (II) – 2

- Learn to use functions and formulas.
 - Learn basic math functions including:
 - [Sum](#)
 - [Subtract](#)
 - [Multiply](#)
 - [Divide](#)
 - [Count](#)

<https://www.youtube.com/watch?v=Jl0Qk63z2ZY>



Basic Excel (II) - 3

- Learn to use functions and formulas.
- Learn basic statistical functions including:
 - [AVERAGE](#)
 - [MAX](#)
 - [MIN](#)
 - [MEDIAN](#)
 - [MODE](#)
 - [STDEV.S](#)
 - [QUARTILE.INC](#)
 - [PERCENTILE.INC](#)



Please download the **penguins** Excel file from Canvas



Basic Excel (II) – 4

- How to install an Add-in -> Data Analysis

- [Instructions](#)

Load the Analysis ToolPak in Excel

Excel for Microsoft 365, Excel for Microsoft 365 for Mac, Excel 2021, [More...](#)

If you need to develop complex statistical or engineering analyses, you can save steps and time by using the Analysis ToolPak. You provide the data and parameters for each analysis, and the tool uses the appropriate statistical or engineering macro functions to calculate and display the results in an output table. Some tools generate charts in addition to output tables.

The data analysis functions can be used on only one worksheet at a time. When you perform data analysis on grouped worksheets, results will appear on the first worksheet and empty formatted tables will appear on the remaining worksheets. To perform data analysis on the remainder of the worksheets, recalculate the analysis tool for each worksheet.

Was this information helpful?

Yes

No



Basic Excel (II) – 5

- Descriptive Statistics using the Data Analysis add-in

Load the Analysis ToolPak in Excel

Excel for Microsoft 365, Excel for Microsoft 365 for Mac, Excel 2021, [More...](#)

If you need to develop complex statistical or engineering analyses, you can save steps and time by using the Analysis ToolPak. You provide the data and parameters for each analysis, and the tool uses the appropriate statistical or engineering macro functions to calculate and display the results in an output table. Some tools generate charts in addition to output tables.

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Was this information helpful?

Yes

No

https://www.youtube.com/watch?v=UZdEWo_a4Wc





Basic Excel (II) - 6

- Learn to use functions and formulas.
 - Learn to use [SUMIF](#) / [COUNTIF](#) and [SUMIFS](#) / [COUNTIFS](#) to add cells only when certain conditions are met.

<https://www.youtube.com/watch?v=BO1-prCG8DI>





Basic Excel (II) - 7

- Learn to use functions and formulas.
 - Learn to use [MAXIFS](#), [MINIFS](#), [AVERAGEIF](#)/[AVERAGEIFS](#)

<https://www.youtube.com/watch?v=EgluXBBtwZ0>

<https://www.youtube.com/watch?v=XZwvyyg6J3M>





Basic Excel (II) - 8

- Learn to use functions and formulas.
 - Nesting Excel functions
- Comment on customized lists

<https://support.microsoft.com/es-es/office/crear-una-lista-desplegable-7693307a-59ef-400a-b769-c5402dce407b>

<https://www.youtube.com/watch?v=Ytc0WZPpA6E>





Basic Excel (II) - 9

- Learn to use functions and formulas.
 - Learn to build standalone logical IF/IFS functions
 - make them more complex by nesting AND and OR within them.

<https://www.youtube.com/watch?v=H3jOF17Klz4>
<https://www.youtube.com/watch?v=KkTaQ5OjAGc>





Basic Excel (II) - 10

- Learn to use functions and formulas.
 - Learn to use the [SUBTOTAL](#) function
 - Use with [Table Filters](#)
 - Use with [Slicers](#)

<https://www.youtube.com/watch?v=0bdhy0ty7Mc>





Basic Excel (II) - 11

- Learn to use functions and formulas.
 1. How to use [STOCKHISTORY](#) function to obtain stock data.

<https://www.youtube.com/watch?v=dQ9wMNVjfCA>





Basic Excel (II) - 12

- Learn to use functions and formulas.
 - Learn to break apart text with the [LEFT](#), [RIGHT](#), [MID](#), [FIND](#) and [LEN](#) functions.

<https://www.youtube.com/watch?v=rbUYH5n0DuM>





New Text Features

- Textafter and Textbefore
- Repeat the previous exercise with these features

<https://www.youtube.com/watch?v=p92Ta3J5vGU>



Basic Excel (II) - 13

- Learn to use functions and formulas.
 - Learn to use [XLOOKUP](#) to find an approximate/exact match and return it or the corresponding value from another column.
([VLOOKUP](#)/[HLOOKUP](#)/ are discontinued)

<https://www.youtube.com/watch?v=4c0CLUER6nw>



Basic Excel (II) – 14

- Time-related formulas

- [Now](#)
- [Today](#)
- [Month](#)
- [Year](#)
- [Day](#)
- [Days](#)
- [Weekday](#)
- [Yearfrac](#)

<https://www.youtube.com/watch?v=vXj0J5MgPTs>





Basic Excel (II) - 15

- Unique Registers [UNIQUE\(\)](#)

<https://www.youtube.com/watch?v=oW-lpQW1MpQ>



Basic Excel (II) - 16

- Learn to use functions and formulas.
 - Learn to work with formula auditing tools including [tracing precedent and dependent cells](#).

<https://www.youtube.com/watch?v=TgT2jeP-TFw>

